

Scientific Reviewer

12+3 · accept-with-notes framing

Strengths: causal L_ctx story · reports trade-offs · fair AC reuse · NYU vs DINOv3 protocol. Weaknesses: n=10 · +20% mixes horizons · ViT-g under VJ2 until G · ADE≠SOTA · cool-down confound.

Shared spine — tattoo every deck

Six claims that must land in every seat

1 Unlock dense features in the JEPA world-model line — L_ctx + deep SS + VisionMix/ViT-G.

2 CLEAR lineage: I-JEPA → V-JEPA → V-JEPA 2 → V-JEPA 2.1 → 2-AC.

3 Franka grasp 60→80%; latent nav ~10× cheaper than NWM.

4 Triangle: SigLIP/DINO | MAE | JEPA WM (now dense).

5 Must: NYU 0.307; ADE 22.2→47.9; Ego4D 7.71; Tables 6-7; L_ctx alone hurts globals / deep SS restores.

6 SSv2 77.7; EK100 40.8; Tartan ATE 5.687 / 10.6s vs NWM 103.2s.

Strengths — what holds up under review

Causal story + honesty about trade-offs

Causal L_ctx story

Hypothesis (context-as-register) → ablation → PCA + linear probes → recipe stack (Fig 3/5, T1).

Reports trade-offs

L_ctx alone nukes SSv2 (−10 pts) — paper shows the hit and the recovery via deep SS.

Fair AC reuse

Public AC codebase + same task configs as V-JEPA 2 — fairer than reinvented planner.

NYU protocol

Dense eval matched to DINOv3 (Siméoni) — credible 0.307 vs 0.309 claim.

Distillation not vanity

ViT-L nearly closes gap to G (T11) — 2B is not a pure scale flex.

Fig. 3 — smoking-gun ablation visual

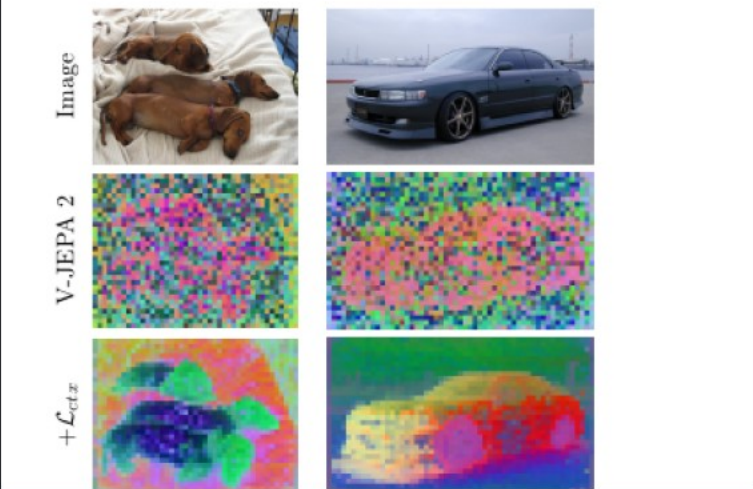
Reviewer loves isolatable causal evidence

erized with Vision Transformer (Dosovitskiy, 2020) and we (2025). Refer to Appendix A for more details.

Vision Tasks

ach for understanding global semantic information from ch specific goals (Bardes et al., 2024; Assran et al., 2025), of V-JEPA 2 features for dense vision tasks. To address through qualitative visualizations and dense downstream

Principal Component Analysis (PCA) of patch features p the first three components to the RGB color channels. using a linear probing protocol, in which we train a single e evaluate V-JEPA 2 on semantic segmentation using the stimation on NYUv2 (Silberman et al., 2012b). Refer to p.



Ask: is distance-weighted λ uniquely necessary vs any dense aux? Table 2 helps.

Fig. 5 / Table 1 — recipe is stacked not factorial

Strength: shows Pareto path. Weakness: confounded with scale.

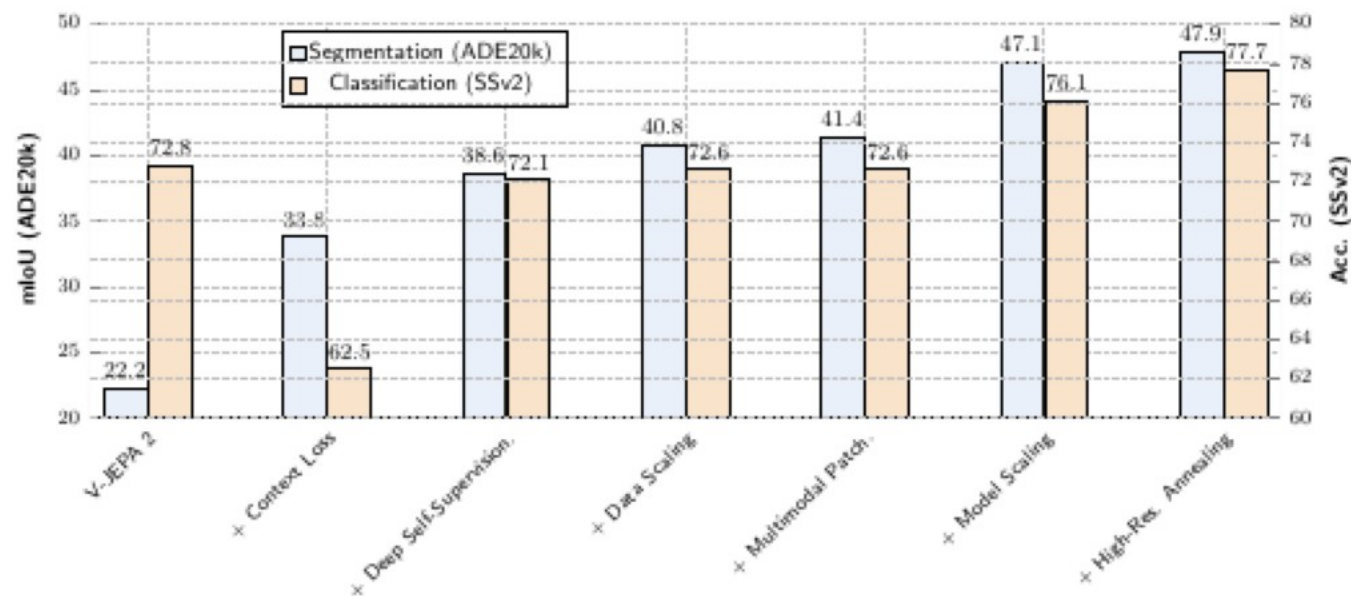


Figure 5 Impact of individual components of our novel V-JEPA 2.1 training recipe. The ablation is conducted starting from a ViT-L architecture, on single-image semantic segmentation on ADE20k, and action classification on SSv2. Introducing *Weighted-Context Self-Supervision* with the context loss significantly improves segmentation, at the cost of classification. *Deep Self-Supervision* allows to retrieve the performance, and further improving segmentation. Scaling the image data with VisionMix 163M dataset, combined with a *Multi-Modal Tokenizer* further improve the results. Finally, the recipe *scales* with model size and high resolution *cool-down*.

Demand: isolate L_ctx / deep SS on fixed tiny data (Empiricist E1-E3).

Weaknesses — ask-for-revision style

Five pressure points

n=10 robot

No CIs / multi-object / multi-env. Don't oversell grasp 80%.

+20% mixes horizons

Matched H=1 is +10 pp (60→70); +20 pp needs H=8. Clarify abstract.

ViT-g under VJ2

SSv2 76.9 vs 77.3; EK Action 38.4 vs 39.7 — gains need ViT-G. Say so loudly.

ADE ≠ SOTA

47.9 vs DINOv3-7B 55.9. 'SOTA dense' overstated outside depth.

Cool-down confound

Resolution ↑ mixed with LR decay — ablate separately?

Nav 10x — impressive but joint system claim

Repr change + sampler change (8 vs ≥ 128 DDIM)

Pressure

Table 7: 10.6s vs NWM 103.2s with competitive Tartan ATE 5.687. Ensure quality parity isn't just easier search in smoother latent space. Different latent dim vs SD-VAE. Still CLEAR in-paper number — footnote the confound.

Allowed phrasing: '~10x faster planning under CDiT+8-DDIM on VJ2.1 features vs SD-VAE NWM.' Forbidden: 'JEPA is 10x better than diffusion WMs' without caveats.

Internal checklist before citing

Reuse discipline

☐ Reproduce Table 1 stages on ViT-L with public data-mix subset (or cite Empiricist smoke direction).

☐ Matched-horizon-only Franka table before citing +20%.

☐ Probe V-JEPA 2.1 vs DINOv3 on BEHAVIOR-relevant affordance / contact metrics.

☐ When citing planning recipe origins, cite V-JEPA 2 / 2-AC + this encoder swap.

☐ Footnote dense=linear (DINOv3-style) vs global=attentive (VJ2-style) when comparing families.

☐ Don't ship context-loss-only checkpoints — globals regress until deep SS.

Verdict

Accept landmark dense unlock · WITH NOTES

ACCEPT — landmark contribution

Dense predictive loss + deep SS is a real, causal, well-documented unlock for video JEPAs. NYU linear SOTA-ish vs DINOv3-7B is the headline that sticks.

WITH NOTES

Clarify +20% grasp abstract; isolate cool-down; don't overclaim ADE SOTA; footnote nav system confound; insist ViT-G for some global wins.

Fig. 8 — robotics fairness check

Same planner configs → attribution to encoder is credible

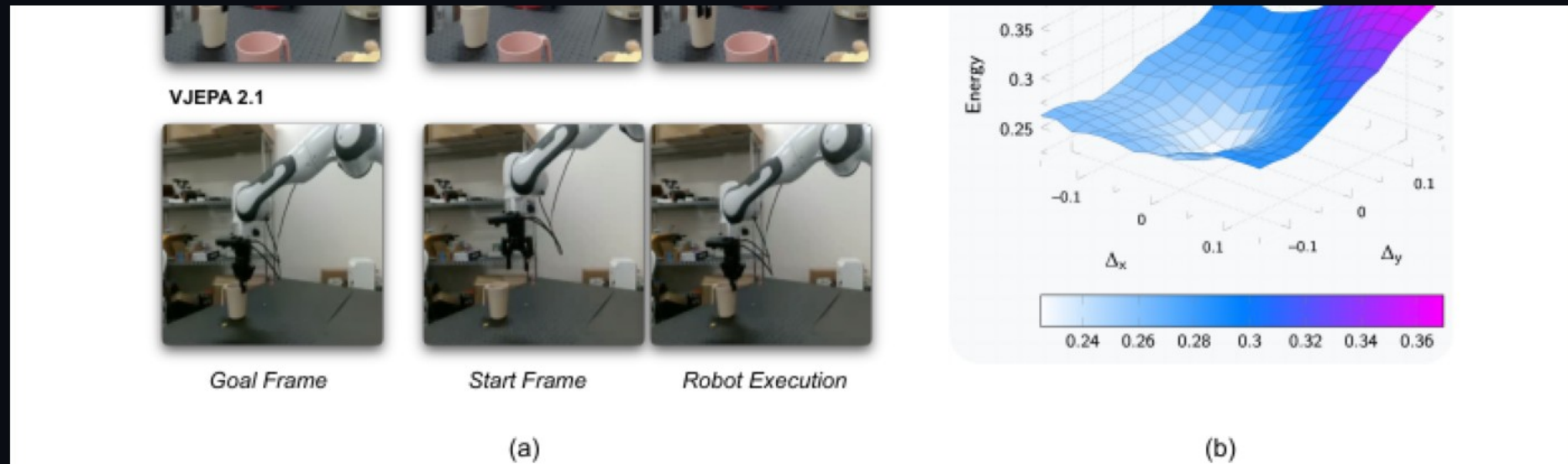


Figure 8 Zero-shot evaluation of VJEPA 2 and VJEPA 2.1 models on a table-top Franka Panda robot arm with a parallel-jaw gripper, demonstrating *Grasp* of a cup using visual goal specification via model-predictive control. VJEPA 2.1 features better encode depth information, leading to an improvement in robot manipulation when grasping the cup involves reasoning over actions along the depth axis of the camera.

control. We use the same exact task configuration files provided in [Assran et al. \(2025\)](#), and similar planning hyper-parameters.

Success rates are reported in Table 6. VJEPA 2.1 exhibits an improved depth understanding, leading to a 10% improvement in the success rate of *Grasp* compared to VJEPA 2 (cf. Figure 8). Moreover, we find that the VJEPA 2.1 model unlocks the benefit of planning with slightly longer rollouts, perhaps due to the improvement in dense features afforded by our model. In particular, by reducing the number of CEM samples and planning

Still: $n=10$; demand matched-horizon before +20% slogan.

Numbers tattoo — pin these

Reconciled with KANTA_PACK · shared across seats

0.307

NYUv2 RMSE linear
beats DINOv3-7B 0.309

22.2→47.9

ADE path T1
33.8→38.6→47.9

7.71

Ego4D STA mAP
ViT-G

40.8

EK100 Action R@5
ViT-G

77.7%

SSv2 attentive
72.8→62.5→72.1→77.7

60/70/80

Franka grasp n=10
matched→H=8 CEM

5.687

Tartan Drive ATE
10.6s vs NWM 103.2s

~10×

latent nav cheaper
than NWM SD-VAE

L_ctx

alone hurts globals
deep SS restores

33.8/62.5

T2 weighted $\lambda=0.5$
ADE / SSv2 trade-off

ViT-G 2B

VisionMix163M
+ cool-down 64@384

distill B/L

ViT-L $\approx$ G on
many probes (T11)

Scientific Reviewer — close

Takeaways + exam bite

1. Strengths: causal L_ctx; honest trade-offs; fair AC reuse; NYU protocol; distill not vanity.
2. Weaknesses: n=10; +20% mixes horizons; ViT-g under VJ2 until G; ADE≠SOTA; cool-down confound.
3. Nav 10× is joint system claim — footnote latent+sampler.
4. Verdict: ACCEPT landmark dense unlock WITH NOTES.

EXAM BITE

Accept V-JEPA 2.1 as the dense unlock for video JEPA; require matched-horizon Franka and cool-down isolation before slogans.

Exam bite — Reviewer

30-second critique

The paper's causal story for L_ctx is strong and it honestly reports the global/dense trade-off. Accept the dense unlock. Push back on abstract +20% grasp (mixes horizons), ADE-as-SOTA, ViT-g regressions until ViT-G, cool-down confounds, and nav 10× as a joint system claim. n=10 Franka is promising, not conclusive.